



Service Bulletin Number: 5659817

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Extended Service Interval Program for Locomotive Applications

Extended Service Interval Program for Locomotive Applications

Product Affected

- QSK38 CM850 MCRS
- QSK38 CM2150 MCRS
- QSK50 CM850 MCRS
- QSK50 CM2150 MCRS
- QSK60 CM500
- QSK60 CM850 MCRS
- QSK60 CM2150 MCRS
- QSK78 CM500

Note : If an engine model or an application is not listed above, contact Cummins Distributor or Cummins Filtration.

Content Overview

The purpose of this document is to identify the steps required to extend the service interval for Locomotive applications.

Section 1: Site Assessment

Section 2: Fluid Specifications

Section 3: Fluid and Filter Monitoring

Section 4: Initial Equipment Upgrades and Filter Requirements

Section 5: Extended Maintenance Intervals

Section 6: Lubricating Oil Drain Interval Extension Test

Section 7: Additional Service Literature

Maintenance intervals discussed within this document include:

- Coolant drain
- Coolant filters
- Fuel filters
- Lubricating oil drain
- Lubricating oil filters

Note : The guidelines and requirements in this bulletin are subject to change without notice. Consult your local Cummins Distributor or Cummins Filtration.

Note : Before adopting this information, it is the responsibility of the end user to contact Cummins Distributor or Cummins Filtration to understand the information in its entirety. Cummins Inc. is not responsible for inaccurate handling of this procedure and failure to Cummins Warranty.

Extended Service Interval for Locomotive Applications Program Requirements

Section 1: Site Assessment

The equipment, site, operating conditions, and environment affect the ability to extend maintenance intervals. The make and model of the equipment, site's rigorous maintenance practice, higher and low loads, long idling periods, and weather and environmental extreme conditions such as wet, dusty, freezing, and hot operation might require more frequent maintenance than specified in the Maintenance Manual.

A site assessment must be completed to ensure success with this extended maintenance interval program. To schedule a site assessment, contact your local Cummins Distributor or Cummins Filtration.

Section 2: Fluid Specifications

Note : Any deviations from the fluid (lubricating oil, fuel, coolant) and filter specifications as defined in this service bulletin will void the use of this extended service bulletin. Fluid and filters not meeting the requirements listed in this document must follow the standard maintenance intervals defined in the corresponding Operation and Maintenance Manual.

Lubricating Oil Requirements

- Lubricating oil **must** meet the requirements specified by Cummins Inc.
 - See Fluids for Cummins Products Service Manual, Bulletin 5411406.
- Lubricating oil used **must** be registered in Cummins Oil Registration List.

Note :

https://quickserve.cummins.com/qs3/qsol/service/serviceproducts/oil_registration.html

- Lubricating oil drain intervals **must** follow the intervals specified in the corresponding Operation and Maintenance Manual unless extended drain intervals are proven through a testing protocol described in this bulletin.
 - For best results, use a fully synthetic oil certified to CES 20078 or CES 20086.
 - Some blended oils or natural base oils can achieve extended service intervals.

Note : During testing it is best **not** to mix two types or brands of oil, even if both are registered to an appropriate Cummins specification. In some cases, it has been observed that mixing oils led to challenges in interpreting the effects of extended oil drains from the mixing of oils.

Fuel Requirements

- Fuels must meet the requirements as specified by Cummins Inc.
 - See Fluids for Cummins® Products Service Manual, Bulletin 5411406.

Specific fuel requirements include:

- Fuels **must** meet International Organization for Standardization (ISO) 4406 cleanliness code of 18/16/13 or better at the inlet of the stage 1 fuel filter with a maximum water content of 200 ppm.
- Fuels that meet or exceed American Society for Testing and Materials (ASTM) D2068 "Filter Plugging Tendency Tests" procedure A with a Filter Blocking Tendency (FBT) calculation high limit of 1.41.
- Fuel **must** pass the fuel quality test using the Cummins Fleetguard Portable Fuel Cleanliness Analysis Kit, part number FK36000.
- Fuels that meet or exceed ASTM D975 - 15b "Standard Specification for Diesel Fuel Oils."
- Only fuel additives that have demonstrated no negative effect on fuel filter or fuel system life can be used.
- If winter, cold weather, or high altitude fuel is used:
 - The applicability of this service bulletin is limited to fuels not exceeding 60% kerosene / 40% diesel content.
- If biofuel is used:
 - The applicability of this service bulletin is limited to fuels not exceeding B5 or 5% biofuel content.
- If sulfur content in fuel is higher than 15 ppm, lubricating oil drain intervals **must** be verified using the procedure in Section 6.

- If sulfur content in fuel is higher than 500 ppm, extended drain intervals will not be supported.

Coolant Requirements

Note : Do not mix two types of coolant products or brands.

Note : Do not top off coolant system with water or conventional coolant.

- Coolant **must** meet the requirements specified by Cummins Inc.
 - See Fluids for Cummins® Products Service Manual, Bulletin 5411406.
- Coolants used **must** be registered in Cummins Coolant Registration List and in accordance with CES 14603 coolant specification and be fully formulated pre-mixed organic acid technology (OAT) coolant. Reference Coolant Registration List.

Note :

https://quickserve.cummins.com/qs3/qsollservice/serviceproducts/coolant_registration.html

- The following coolants have been proven capable of meeting extended drain intervals:
 - Fleetguard® ES Compleat™ EG OAT Coolant
 - Fleetguard® PG Platinum™ Heavy Duty Engine Coolant
 - Cummins® TEC Infinity™ Extended Life Coolant
 - Cummins® PGXL™ Vital Advanced Extended Life Heavy Duty Coolant

Section 3: Fluid and Filter Monitoring

This section provides the guidelines and requirements to collect, monitor, flag, act, and analyze engine fluids and filters that provide information to guide the Extended Service Interval for Locomotive Applications Program.

Lubricating Oil Sampling and Analysis:

- See Service Bulletin, Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060.
 - Elements to Monitor for High Horse Power Engine Oil Analysis
 - Oil Sample Collection
 - Setting Oil Analysis Flag Limits
 - Investigation and Corrective Action Based on Flag Limits
- See Service Bulletin, Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452.

Fuel Sampling and Analysis:

- For High Pressure Injection (HPI) Engine, Cummins Inc. recommends fuel samples be taken directly from vehicle equipment at the inlet to suction side fuel filter.
- For Modular Common Rail System (MCRS) Engines, samples **must** refer to Bulletin 4022123, Fuel Sampling for Particle Counting Test.
- If the fuel sample does **not** meet the requirements specified, revisit the site audit to identify opportunities for improved fuel quality.
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Coolant Sampling Analysis:

- See the corresponding Operation and Maintenance Manual, under Section V - Maintenance and Specification, procedure 018-004 Coolant Recommendations and Specifications
- If molybdate, nitrite, and pH are **not** within specification, investigate maintenance practices and send coolant sample for contamination check. Coolant **must** be replaced. See the corresponding Service Manual. Reference procedure 008-018 Cooling System.
- If molybdate, nitrite, and pH are **not** within specification, investigate maintenance practices and send coolant sample for contamination check. Coolant **must** be replaced. See the corresponding Service Manual.

Table 1 and Table 2 below specify the fluid analysis to be performed during operation when using extended service intervals.

Table 1, Engine Fluid Sampling Analysis Requirements

Fluid	Sampling Interval	Suggested Method and Location	Purpose and Test Package
Lubricating Oil	250 hours	Per Service Bulletin 4022060: Collect oil sample from pressurized port while engine idling or by vacuum pump through lubricating oil dipstick retaining tube.	Test1 for Oil Quality and Contaminants:
			• Elemental Metals by Inductively Coupled Plasma (ICP)
			• Viscosity at 100°C [212°F]
			• Water Percentage
			• Fuel Dilution
			• Soot Percentage
			• Oxidation I Nitration
Coolant	4000 hours	By vacuum pump through radiator fill cap or through quick fill I drain port	Test2 for Coolant Replacement Limits:
			• Sulfate
			• Chloride, Fluoride, Bromide
			• pH
			• Oil and fuel contaminants
			• Grease, solder bloom, silica gel, rust or scale
Coolant	Every 6 months	By vacuum pump through radiator fill cap or through quick fill I drain port	Test for Glycol Levels:
			• Coolant Test Kit, Fleetguard Part Number CC8997
Fuel	As Needed: Sample fuel from equipment for quality and cleanliness if fuel filters do not reach required life or fuel filter life decreases from established baseline.	Use sampling port upstream of Stage 1 fuel filters and at bottom of equipment fuel tank through port on equipment fuel tank.	Reference the Fuel Filter Plugging Bulletin, 5504538.

If using Cummins® Fleetguard® Monitor: 1Reference Monitor™ CC36136, 2Reference Monitor™ CC2700.

Table 2, Bulk Storage, Tank Storage, and Service Equipment Fluid Sampling Requirements

Fluid	Sampling Interval	Suggested Method and Location	Purpose and Tests Cummins® Monitor Program Test Package
Bulk Storage - Lubricating Oil	Minimum of quarterly and if change to lubricating oil brand or specification	By sampling valve installed in bulk tank	Test1 for Oil Quality and Contaminants:
			• Elemental Metals by Inductively Coupled Plasma (ICP)
			• Viscosity at 100°C [212°F]
			• Water Percent
			• Fuel Dilution
			• Soot Percent
			• Oxidation I Nitration
			• Total Base Number (TBN)
Bulk Storage - Coolant	Minimum of quarterly and if change to coolant brand or specification	By sampling valve installed in bulk tank	Test2 for Coolant Quality and Contaminants:
			• Visuals (foam, color, oil, fuel, magnetic and non- magnetic precipitate and odor)
			• 12 Metals by Inductively Coupled Plasma (ICP)
			• Percent Antifreeze
			• Freeze Point
			• Boil Point
			• pH
			• Hardness
			• SCA Number
			• Nitrites
			• Total Dissolved Solids
			• Specific Conductance

Bulk, Tank Storage and Field Service Equipment - Fuel	Monthly or more frequently if fuel filters of equipment do not reach required life or filter life decreases from established baseline.	By sampling valve installed in bulk tank, with particle counter in place and sample bottles. At bottom of equipment fuel tank through port on equipment fuel tank with sample bottles.	Test3 for Fuel Quality and Contaminants:
			• 24 Metals by Inductively Coupled Plasma (ICP)
			• Viscosity at 40°C [104°F]
			• Percent Sulfur
			• API Gravity
			• Distillation
			• Flash Point
			• Cloud Point
			• Pour Point
			• Water and Sediment
			• Bacteria, Fungi and Mold
			• Calculated Cetane Index
			• Thermal Stability
			• Water by Karl Fischer titration method
			Test4 for Fuel ISO Particle Counts and Contaminants:
			• PPM ISO Particle Count
			Test for Filter Plugging:
			• Cummins Fleetguard Portable Fuel Cleanliness Analysis Kit, part number FK36000 or ASTM D2068 - Procedure A.
If using Cummins® Fleetguard® Monitor™: 1Reference Monitor™ CC36136 2Reference Monitor™ CC2700 3Reference Monitor™ CC2651, 4Reference Monitor™ CC2719			
For more information about Fleetguard® Monitor™, reference CumminsFiltration.com.			

Section 4: Initial Equipment Upgrades and Ongoing Filter Requirements

Note : The guidelines and requirements in this bulletin are subject to change without notice. Consult your local Cummins Distributor or Cummins Filtration.

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Note : Failure that might occur due to installing or using other manufacturer's auxiliary devices, accessories or consumables, are not covered under Cummins® Warranty.

Note : Any deviations from the fluid and filter specifications as defined in this service bulletin will void the use of this extended service bulletin. Fluid and filters not meeting the requirements listed in this document **must** follow the standard maintenance intervals defined in the corresponding Operation and Maintenance Manual.

Extending maintenance intervals requires the following:

- Fuel: Advanced filtration inclusive Industrial Pro™ fuel systems with Nanonet® media
- Lubricating Oil: Venturi™ Combo lube filters and/or Spiratec™ centrifuge lubricating oil filtration
 - Spin-on filters: Venturi™ Combo, Stratapore™, Nononet™ Combo, and/or
 - Eliminator: Spiratec™ centrifuge lubricating oil filtration
- Coolant: Stratapore™ extended service coolant filtration

Lubricating Oil Filters

For engines with Eliminator™, replace Eliminator™ rotor with one-piece centrifugal lubricating oil filter with SpiraTec™ technology as shown in Table 3. Reference Technical Service Bulletin 160036.

Perform the following checks:

1. The casting tolerance of the centrifuge cover has been adjusted to accommodate the new centrifugal filter. Centrifuge covers with the adjusted tolerance are stamped with the letter "V". If the centrifuge cover is not stamped with the letter "V", replace the centrifuge cover with a centrifuge cover, Part Number 4016423, stamped with the letter "V" or grind/file any high spots found at the parting line on the inside of the centrifuge cover.
2. Measure clearance between the spindle and the bushings inside the one-piece centrifugal lubricating oil filter, replace the spindle if out of specification. Reference Bulletin 3666253.

Table 3, Centrifuge Oil Filter Part Numbers for Eliminator™ or Remote Centrifuge

Engine Model	Description	Cummins® Part Numbers	Fleetguard® Part Number
QSK50 (All Versions) QSK60 (All Versions)	One-Piece Centrifugal Oil Filter with SpiraTec™ Technology	4331734	CS41018
QSK78 (All Versions)	One-Piece Centrifugal Oil Filter with SpiraTec™ Technology	3649569	CS41019

Fuel Filters

For engines with a High Pressure Injection (HPI)® fuel system, remove the on-engine fuel filter head and install off engine Industrial Pro fuel filter as shown in Table 4. Reference installation instructions included in the box with the fuel filters.

For engines with a Modular Common Rail System (MCRS) fuel system, upgrade stage 1 and stage 2 filtration as shown in Table 4. Reference installation instructions included in the box with the fuel filters.

Table 4, Fuel Filter Head and Fuel Filter Element Part Numbers

Engine Model	Filtration Stage	Description	Quantity	Cummins Part Number	Fleetguard Part Number	Quantity	Cummins Part Number	Fleetguard Part Number
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QSK38 (All Versions) QSK50 (All Versions) QSK60 (All Versions)	Stage 0 (only required if Cummin s lift pump is used)	14" Assembl y, Drain Valve, 25 Micron filter	1	3639554	Not Applicabl e	1	3639672	Not Applicabl e
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QSK38 (All Versions)	Stage 1	Industrial Pro Unit, Single, Tall, Bracket, WIF, Clear Bowl with 5 Micron NanoNet ® filter. In Left / Out Right	1	4328042	FH23901	1	4328043	FS53014
		Industrial Pro Unit, Single, Tall, Bracket, WIF, Clear Bowl with 5 Micron NanoNet ® filter. In Right / Out Left	1	4328041	FH23903	1	4328043	FS53014
		Industrial Pro Unit, Single, Tall, Bracket, WIF, Clear Bowl with 5 Micron NanoNet ® filter. In Left / Out Left	1	4328064	FH23905	1	4328043	FS53014

		Industrial Pro Unit, Single, Tall, Bracket, WIF, Clear Bowl with 5 Micron NanoNet ® filter. In Right / Out Right	1	4328065	FH23906	1	4328043	FS53014
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QSK50 (All Versions) QSK60 (All Versions)	Stage 1	Industrial Pro Unit, Dual Tall, Bracket, WIF, Clear Bowl with 5 Micron Nanonet filter. In Right / Out Right	1	4381962	FH23907	2	4328043	FS53014
		Industrial Pro Unit, Dual Tall, Bracket, WIF, Clear Bowl with 5 Micron Nanonet filter. In Right / Out Left	1	4381959	FH23908	2	4328043	FS53014
		Industrial Pro Unit, Dual Tall, Bracket, WIF, Clear Bowl with 5 Micron Nanonet filter. In Left / Out Right	1	4381960	FH23910	2	4328043	FS53014

		Industrial Pro Unit, Dual Tall, Bracket, WIF, Clear Bowl with 5 Micron Nanonet filter. In Left / Out Left	1	4381961	FH23911	2	4328043	FS53014
QSK78 (All Versions)	Stage 1	Industrial Pro Unit, Triple Tall Tall, Bracket, WIF, Clear Bowl with 5 Micron Nanonet filter. In Left / Out Right	1	5374777	FH23930	3	4328043	FS53014
QSK38 CM850 CM2150	Stage 2	On Engine or Remote Mount Configuration Filter Head with 3 Micron NanoNet ® filter	1	4380732	Not Applicable	2	5375819	FF63042 NN
QSK50 CM850 CM2150 QSK60 CM850 CM2150	Stage 2	On Engine or Remote Mount Configuration Filter Head with 3 Micron NanoNet ® filter	1	4380859	Not Applicable	3	5375819	FF63042 NN

If an application uses a MCRS CM850 or MCRS CM2150 for Tier 2 and needs to be upgraded to Tier 4 Stage 2 Filtration Package, the two-stage air bleed valve is needed as shown in Table 5. Reference Technical Service Bulletin 1800021.

Table 5, Associated Hardware

Description	Cummins® Part Number
2 Stage Air Bleed Valve	4379736

Note : Air bleed valve is only needed when upgrading fuel filter head.

Coolant Filters

The upgraded coolant filtration options offer higher capacity and improved filtration performance. Remove existing coolant filters and replace with StrataPore™ coolant filters. See Table 6 below for part numbers.

Table 6, Coolant Filter Part Numbers

Description	Cummins® Part Number	Fleetguard® Part Number
On-engine, no DCA*, StrataPore™ media	3098690	WF2123
*Only filters with no DCA should be used in combination with OAT coolant.		

Section 5: Extended Maintenance Intervals

Note : Refer to the Service Manual for different ambient temperature engine start up conditions. Refer to Fluid Cummins Product for change of elements on different ambient conditions.

Seasonal Requirements:

Cold Weather Preparation:

- Check operation of fuel filter heaters.
- Test for Glycol Levels with Coolant Test Kit, Fleetguard Part Number CC8997.
- Check coolant level / Top Off. See equipment manufacturer service information.

Note : Extreme winter or salt corrosion environments may require lubrication of the collar threads on stage 1 fuel filters with anti-seize compound, Part Number 3163086, every 180 days.

Hot Weather Preparation:

- Test for Glycol Levels with Coolant Test Kit, Fleetguard Part Number CC8997.
- Check coolant level / top off. See equipment manufacturer service information.

Time and Condition Based Requirements:

Table 7, Time Based Maintenance Intervals for Locomotives

Syste m	List of Maint enanc e Tasks	As Requi red	Daily and Weekl y	250 Hours	500 Hours	1000 Hours	1500 Hours	2000 Hours	3000 Hours	4000 Hours	4500 Hours
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Coolant	Test coolant for replacement limits	X								X	
Coolant	Change coolant filters									X	
Coolant	Top off coolant	X			X	X	X	X	X	X	X
Fuel	Drain water and sediment from lower bowl of Industrial Pro™ fuel filters	X	X	X	X	X	X	X	X	X	X
Fuel	Replace stage 1 fuel filters	X					X		X		X
Fuel	Replace stage 2 fuel filters	X					X		X		X

Lubricating Oil	Collect lubricating oil sample and perform lubricating oil analysis	X		X	X	X	X	X	X	X	X
Lubricating Oil	Change engine lubricating oil		According to Operation and Maintenance Manual or Lubricating Oil Drain Interval Established in Section 6.								
Lubricating Oil	Change rotor in Eliminator™ / Centrifuge						X		X		X

If the QSK38 CM850 CM2150 uses the QSK50/60 options and quantities outlined in Table 4, 2000 hours can be achieved.

Table 8, Condition Based Filter Maintenance Intervals

System	Replacement Indicators
Fuel Filters, Stage 1	Replace based on schedule in Table 7 or using visual indication on Industrial Pro™ units or when indicated by FleetguardFIT™.
Fuel Filters, Stage 2	Replace based on schedule in Table 7 or when indicated by FleetguardFIT™.
Lubricating Oil, Eliminator™ / remote centrifuge rotor	Replace based on schedule in Table 7 or at the same time as lubricating oil and lubricating oil filters are replaced.
Coolant / Water filters	Replace based on schedule in Table 7.

FleetguardFIT™

Filtration Intelligence Technology (FIT) is a real-time filtration monitoring system that provides full visibility to filter (Fuel, Lube and Air) and oil life enabling proactive problem detection using intelligent sensing and advanced data analytics. To learn more about FleetguardFIT and how to get it onto your engine, contact Cummins Filtration at info@fleetguardfit.com.

Note : If fuel filters do not reach required life or filter life decreases, assessment of fuel quality, fuel tank condition, storage systems and bulk filtration system is required.

Section 6: Lubricating Oil Drain Interval Extension Test

⚠ CAUTION ⚠

The use of a synthetic base lubricating oil does not justify extended lubricating oil change intervals. Extended oil change intervals can decrease engine life as the result of corrosion, deposits, and wear.

⚠ CAUTION ⚠

If the equipment is using a remote lubricating oil tank in which lubricating oil is continuously circulated (not a Centinel™ oil make-up tank), the lubricating oil in the remote tank must be changed when the lubricating oil in the engine lubricating oil sump is changed.

The purpose of this test is to increase the interval between lubricating oil changes using lubricating oil analysis results from the Fleetguard® Monitor Fluid Analysis Program or another equivalent laboratory.

This method for setting the Lubricating Oil Drain Interval with Filter Optimization replaces the use of the Lubricating Oil Chart Method when using this Service Bulletin.

Do **not** extend lubricating oil and lubricating oil filter change intervals beyond the limits specified in the maintenance schedule, unless all conditions required by this service bulletin are met:

- Completion of site assessment as defined in Section 1.
- Lubricating oil meets Specifications in Section 2.
- Use of lubricating oil filters specified in Section 3.
- Lubricating oil monitoring system is in place as specified in Section 4.

If fleet is made up of mixed engine models, varying duty cycles, or equipment with differing lubricating oil system configurations, each group must be considered separately. A minimum of three units of each should be evaluated thoroughly prior to adopting a drain interval for the fleet. Segment engines by:

- Engine Model
- Lubricating oil system capacity (including any remote lubricating oil tanks where the lubricating oil is continuously circulated)
- Use of Centinel™ lubricating oil replenishment systems
- Use of Eliminator™ vs spin on lubricating oil filters and centrifuge
- Duty cycle
- Lubricating oil used (brand name, performance category, and viscosity grade)

Note : The Lubricating Oil Drain Interval Extension established by this process can only be applied to engines in the same segment as defined above and with the specific lubricating oil used to establish the extended lubricating oil drain interval. Any changes to the lubricating oil (including changes to viscosity grade), duty cycle, or engine lubricating oil system will require re-verification of the lubricating oil drain interval.

Lubricating oil samples must be sent to a lab capable of testing the parameters in Table 9 using the specified test method.

Table 9, Oil Analysis Test Parameters

Parameter Type	Parameter, Unit	Recommended Test Method
Lubricating Oil Health	Total Base Number (TBN), mg KOH/g	ASTM D4739
	Total Acid Number (TAN), mg KOH/g	ASTM D664
	Oxidation, abs/cm	ASTM E2412 or ASTM D7414*
	Nitration, abs/cm	ASTM E2412
	Viscosity at 100°C [212°F], cSt	ASTM D445
Contamination	Fuel Dilution, massPercent fuel in oil	ASTM D7593
	Soot, massPercent soot in oil	ASTM E1131
	Sodium - Na, ppm	D5185
	Potassium - K, ppm	D5185
	Silicon - Si, ppm	D5185
	Aluminum - Al, ppm	D5185
Wear Metals	Iron - Fe, ppm	D5185
	Lead - Pb, ppm	D5185
	Copper - Cu, ppm	D5185
	Aluminum - Al, ppm	D5185
	Tin - Sn, ppm	D5185
	Chromium - Cr, ppm	D5185

*The Fourier transform infrared spectroscopy (FTIR) data for oxidation can use either ASTM method but should report whether trending or subtraction methods were used.

Collection of Lubricating Oil Sample:

Samples recommended to be taken at intervals specified in Table 11. For sampling procedure, See Service Bulletin, Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060.

A minimum of five lubricating oil samples per oil drain interval **must** be collected to establish trend lines. For lubricating oil that can **not** reach the interval established in the sampling plan, collect lubricating oil samples at more frequent intervals (every 50 hours or every 75 hours) to determine the appropriate lubricating oil drain interval.

Label the sample with the following information before submitting for analysis:

- Engine model/serial number
- Miles/hours of lubricating oil use
- Miles/hours accumulated on engine since new or rebuild
- Lubricating oil used (brand name, performance category and viscosity grade)
- Date sample was collected
- Engine application
- Amount of new lubricating oil added since previous lubricating oil change.

Verification of Lubricating Oil Integrity:

Both trending and limits are used to determine the appropriate lubricating oil drain interval. Cummins Inc. and the Customer should collaborate to determine the appropriate lubricating oil drain interval. Documentation of the lubricating oil drain interval validation **must** be retained in the site assessment file.

Linear trending throughout each lubricating oil drain interval of:

- Total Base Number (TBN)
- Total Acid Number (TAN)
- Oxidation
- Nitration
- Soot (Soot may not begin trending immediately; when trending soot is usually linear.)
- Iron
- Lead
- Copper

Linear Trending Example Key for Figure 1:

1. Linear trend of lubricating oil analysis parameter value vs hours on lubricating oil.
2. Point at which lubricating oil analysis parameter value deviates from linear trend line.
3. Hours on lubricating oil at which lubricating oil analysis parameter value deviates from linear trend line.
4. Established lubricating oil drain limit. Subtract 20% from hours on lubricating oil where parameter value deviates from linear trend line.

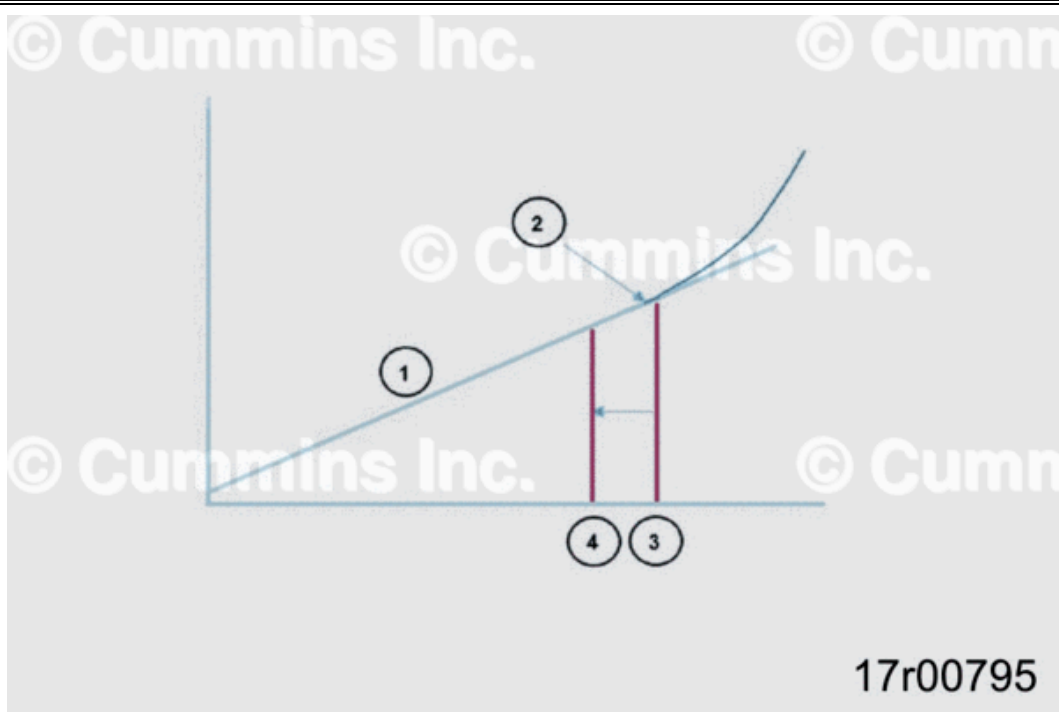


Figure 1, Linear Trending Example

Lubricating Oil Analysis Parameters are within limits defined in Table 10.

Table 10, Oil Analysis Parameter Limits	
Oil Analysis Parameter	Limit Value
Total Acid Number (TAN) mg KOH/g	2.5 increase over new lubricating oil or equal to TBN
Total Base Number (TBN) mg KOH/g	2.5 minimum or equal to TAN
Fuel Dilution Percent	5%
Soot Percent	5%

Table 10, Oil Analysis Parameter Limits	
Oil Analysis Parameter	Limit Value
Viscosity 100°C [212°F] cSt	For XW-40:
	MIN: 12.5 MAX: 16.3
	For XW-30:
	MIN: 9.3 MAX: 12.5
Oxidation abs/cm – direct trending	151
Nitration abs/cm – direct trending	25
1 Increase over new (unused oil sample)	

Repeat sample plan for 3 different lubricating oil drain intervals on each of the 3 engines in the same segment.

Final lubricating oil drain will be the hours at which any of the samples did **not** meet the lubricating oil integrity test criteria reduced by 20%.

20% margin is added to the approved lubricating oil drain interval to account for changes in rate of lubricating oil degradation as the engine ages.

Table 11, Example Lubricating Oil Sampling Plan		
Sample Number	Lubricating Oil Hours	Description
0	Baseline	Sample unused lubricating oil from the container.
1	0	Collect sample after filling the sump with new lubricating oil and letting the engine idle for 5 minutes.
2	250	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
3	500	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
4	750	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
5	1000	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
6	1250	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.

Table 11, Example Lubricating Oil Sampling Plan

Sample Number	Lubricating Oil Hours	Description
7	1300	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
8	1350	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
9	1400	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
10	1450	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
11	1500	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
12	1550	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
13	1600	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
14	1650	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
15	1700	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
16	1750	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
17	1800	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.

Table 11, Example Lubricating Oil Sampling Plan

Sample Number	Lubricating Oil Hours	Description
18	1850	Verify Lubricating Oil Integrity using the steps above. Stop test and change lubricating oil if requirements are not met.
19	1900	End of test. Collect sample before draining lubricating oil.

Section 7: Additional Service Literature

Literature found on QuickServe® Online:

Beginning in 2007, Cummins Inc. started using service model names. For assistance in identifying associated service model name:

- See Technical Service Bulletin, Cummins® Service Engine Model Identification, TSB130080. Review associated Cummins® Product Technology procedures.

Service Bulletins:

- See Service Bulletin, Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060.
- See Service Bulletin, Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452.
- See Service Bulletin, Fuel Sampling for Particle Counting Test, Bulletin 4022123.

Operation and Maintenance Manuals:

- See Cummins® Service Engine Model Identification, Bulletin TSB130080. Review the associated Cummins® Product Technology procedures.
- See QSK45 and QSK60 (MCRS and HPI) Operation and Maintenance, Bulletin 3666260.
- See K38, K50, QSK38 and QSK50 Operation and Maintenance Manual, Bulletin 3810497.
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- Fluids for Cummins Products Service Manual, Bulletin 5411406.
- See Cummins® Service Engine Model Identification, Bulletin TSB130080. Review the associated Cummins® Product Technology procedures.
- See QSK45 and QSK60 (MCRS and HPI) Service Manual, Bulletin 4021530.
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Literature not found on QuickServe® Online:

Hardcopy Service Bulletin:

- Service Bulletin, Fuel Quality for High Horsepower Engines with Modular Common Rail Fuel Systems (MCRS), Bulletin 4091849

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